MILITARY SPECIFICATION

RANGE FINDER, FIRE CONTROL: XM20

1. SCOPE

1.1 This specification covers one type of coincidence image instrument known as Range Finder: XM20 for determining range measurement in meters in the 90MM Gun Tank: M48A3 Fire Control System.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection Of
MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection Of
MIL-P-14232	Parts, Equipment and Tools for Ordnance Material, Packaging Of
MIL-F-14619	Fire Control Parts Equipment and Tools; Packaging Of
MIL-I-45208	Inspection Requirements, General Specification For

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Ordnance Corps

F8205604	Rangefinder, Fire Control: XM20

(Copies of specifications, standards, drawings, and publications required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model.- At the beginning of production one rangefinder, representative of production shall be made available to the Government to determine if the rangefinder meets all requirements of this specification.

3.2 Materials.- Materials shall be in accordance with the drawings, general specifications and material specifications forming part of this specification.

3.3 Construction.- The rangefinder shall be manufactured in accordance with Drawing F8205604 "Rangefinder, Fire Control: XM20" and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.(see 6.4)

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3.4 General specifications.- All requirements as specified in Section 3 of Specification MIL-F-13926 and MIL-O-13830 shall apply, except those requirements not logically related to the rangefinder.

3.5 Pre-vibration.

3.5.1 Orientation.- The rangefinder prior to vibration shall be setup in a horizontal position as shown on the Assembly View Drawing F8205604 and all instructions thereon, and the following conditions shall be established prior to testing.

(1) The line through the centers of the two mounting bearings horizontal.

(2) Identified production end housings in place.

(3) Level bubble centered within width of a vial graduation.

(4) Calibrating reticle illuminated.

(5) Occluder set at mid-index.

(6) Filters out of the optical system.

(7) A target provided at 1200 meters plus or minus 100 meters and the range scale set to agree with the range chosen.

(8) Set the vertical calibration knob at mid-point of total travel within 10 degrees and slip the scale to aline indices. Establish vertical coincidence of the target image.

(9) Set the horizontal calibration knob at mid-point of total travel within one division on the scale. Slip the scale to zero index. Make a series of five horizontal coincidence readings on the target image. Rotate the knob to the median reading.

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(10) Set the deflection (azimuth) and elevation boresight knobs and the auxiliary deflection and elevation boresight knobs at mid-point of total travel within 0.1 mil graduation. Slip the deflection scales to the three (3) position and slip the elevation scales to the two (2) position.

(11) The left boresight reticle shall be set to bring the line of sight 1 mil below the horizontal, and 1 mil to the right of the perpendicular to a center line through the center of the mounting bearing within .05 mil (reticle line width) and the knobs shall then be locked.

(12) The right boresight reticle shall be set so that its line of sight coincides with the aiming point of the left reticle within .05 mil (reticle line width) and the knobs shall then be locked.

3.5.2 Knob travel (calibrated).

3.5.2.1 Horizontal knob.- The horizontal calibration knob shall read within 7 units of error (UOE) of its mid-point of total travel when horizontal coincidence is established at 1200 meters plus or minus 100 meters. If requirement is met, the scale shall then be slipped to the zero index.

3.5.2.2 Vertical knob.- The vertical knob shall be within 30 degrees of its previously established mid-point. If requirement is met slip the scale to aline indices.

3.5.2.3 Knob scales.- The scale readings on all boresight knobs shall be within 0.7 mil (knob graduation) of their mid-points. If requirement is met slip the scales to three (3) position and two (2) position as indicated in 3.5.1 (10).

3.5.3 Calibration reticle setting accuracy.- The calibrating reticle (lower) that is fixed with respect to the external field shall lie in its entirety between 5 and 12 mils above the center of the boresight cross and the vertical line of the calibrating reticle shall be within 3 mils left and 1 mil right of the vertical center line of the sighting reticle. The halving knob shall then be adjusted to position the movable (upper bar) of the calibrating reticle, so that with internal

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correction system (ICS) knob movement, the upper bar will pass within one reticle line thickness of the lower reticle, yet maintain a recognizable separation between the two reticle bars during movement. All coincident readings of the target shall be made in the center five mils of the true field unless otherwise required. Five reticle coincidence settings shall be made with the ICS adjustment. The median of the base five readings shall read within 7 UOE of the (25) position.

3.6 Vibration.- After meeting all the above requirements, the rangefinder shall be vibrated at an amplitude of not less than 1/16 inch (1/8 inch total excursion) at a frequency of 30 cycles per second for a duration of 5 minutes, plus or minus 15 seconds.

3.7 Post-vibration.

3.7.1 Post-vibration optical re-check.- With the rangefinder restored to the conditions of 3.5.1 (1) thru (7), the rangefinder shall meet the requirements of 3.7.1.1 to 3.7.1.4 inclusive. Subsequent to meeting the above requirements the rangefinder shall meet all requirements of 3.7.2 to 3.7.17 inclusive. (see 6.5)

3.7.1.1 Alinement of sighting reticles.- The line of sight of the left telescope through the boresight mark shall not have changed by more than 0.25 mil in deflection and 0.5 mil in elevation from the setting introduced in 3.5.2.3. The line of the sight through the aiming point of the right telescope shall coincide with the aiming point of the left telescope within a maximum angular separation of 0.1 mil.

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3.7.1.2 Image.- The target image shall not nave changed coincidence by more than 0.5 mil in vertical calibration or more than 25 UOE in horizontal calibration as measured against a calibrated external target using the median of five coincident settings made with the horizontal calibration knob. (see 3.9)

3.7.1.3 Coincidence reticle setting.- The space between the calibrating reticles shall not decrease past a distinguishable spacing and shall not increase in excess of 0.2 mil.

3.7.1.4 Horizontal calibration.- Set the horizontal calibration knob at the new median setting (see 3.7.1.2) Then using the ICS knob obtain a new median from five ICS settings of the calibrating reticle. This median shall not differ by more than 5 UOE from the median reading established in 3.5.3 prior to vibration.

3.7.2 Inspection for dirt and bright spots.

3.7.2.1 Inspection for dirt.- The maximum real diameter of any one defect shall not exceed 0.2 mil in the outer 22 - 1/2 degrees, and 0.05 mil in the central 17 - 1/2 degrees of the apparent field as viewed at the eyepiece. The sum of the maximum real diameter of dirt and pinholes in the rangefinder shall not exceed 1 mil. The rangefinder shall meet the cleanliness requirements of Specification MIL-O-13830. Visible dirt on the optical surfaces shall not exceed in size and number, the limitations for size and number of digs permitted for each surface as specified on applicable drawings.

3.7.2.2 Bright spots.- Bright spots (pinholes) in reticle shall not exceed in size and number the limitations for size and number of digs permitted for that surface as specified on the applicable drawings.

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3.7.3 Parallax.

3.7.3.1 Parallax of sighting reticles.- Utilizing a target at 1200 meters plus or minus 100 meters and with the range scale set to agree with the range chosen, parallax in the right and left systems at the optical axis (boresight cross) shall not exceed 0.1 mil, nor shall there be more than 0.1 mil parallax between reticles.

3.7.3.2 Parallax of the calibrating reticles.- The median of five ICS readings taken with the eye centered at 0.030 inch to the left, and the median of five readings taken with the eye centered at 0.030 inch to the right of the center of the common exit pupil shall not differ by more than 6 UOE as measured on the ICS knobs.

3.7.3.3 Parallax of the target images.- Utilizing a target at 1200 meters plus or minus 100 meters the median of five coincidence readings taken with the eye centered 0.030 inch to the left of the center of the exit pupil shall not differ from the median of five coincidence readings taken with the eye centered 0.030 inch to the right of the center of the exit pupil by more than 6 UOE as read on the horizontal calibration knob.

3.7.4 Exit pupil.- The minimum diameter of the common exit pupil when viewed in the direction of the optical axis shall be 0.110 inch (2.79 millimeters).

3.7.5 Eyepiece focus.- The eyepiece focus shall be adjustable over the range of plus or minus 3 diopters as read on the diopter scale. The zero of the diopter scale shall indicate zero diopters within 0.25 diopter as measured against the left target image.

3.7.6 Resolution.- The resolution on the optical axis of each optical system shall be 6 seconds of arc or less.

3.7.7 Light transmission.- The difference in light transmission between the two optical systems shall not exceed 30 percent of the higher transmission value.

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3.7.8 Sighting reticles.

3.7.8.1 Accuracy of sighting reticles.- The angular value of the total horizontal lines and spaces of the reticles, in either field of view shall be 40 mils plus or minus 1.0 mil.

3.7.8.2 Plumb travel.- The left boresight mark when projected into the eyepiece shall not deviate from a plumb line by more than 0.2 mil when the rangefinder is elevated from 8 degrees depression to 19 degrees elevation, starting with the line of sight in coincidence with the plumb line.

3.7.8.2.1 Elevation travel.- The rangefinder shall display unrestricted movement through an excursion of at least 22 degrees in elevation above a horizontal line of sight and 15 degrees in depression below a horizontal line of sight.

3.7.8.3 Parallelism of sighting reticle and image.- When either sighting reticle is projected into the eyepiece the forty mil horizontal reticle line shall be parallel to the image of a horizontal target line within 43 minutes of arc (43 minutes of arc is equivalent to an angle subtended by a 0.5 mil block at one end of the forty mil reticle line when the other end of the reticle is in coincidence with the target line).

3.7.8.4 Tilt variation between reticles.- When the right reticle is superimposed on the left reticle, the variation between the reticles as measured at the 40 mil horizontal reticle lines shall not exceed the width of a reticle line at one extremity of the 40 mil reticle lines when the other extremity of both 40 mil reticle lines are coincident.

3.7.9 Coincidence reticles.

3.7.9.1 Parallelism of calibrating reticles.- Each calibrating reticle shall be parallel to the image of a plumb line within 1 degree of arc.

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3.7.9.2 Reversing reticles.- Utilizing the halving knob, the upper and lower reticle bars shall be reversed by putting the upper bar below. The median of five reticle coincidence readings on the ICS knob shall not differ by more than 5 UOE from the median established in 3.7.1.4.

3.7.9.3 Vertical movement accuracy.- The rangefinder shall be reset to the requirements of 3.7.1.4. When the vertical calibration knob is rotated to its maximum counterclockwise (CCW) position and the reticle returned to its original position by means of the halving knob the median of 5 deflection readings (UOE) of the calibrating reticle using the ICS knob shall not differ by more than 10 UOE from the median of 5 deflection readings obtained when the vertical calibration knob is rotated to maximum clockwise (CW) position and the reticle to its original position by means of the halving knob.

3.7.9.4 Horizontal movement accuracy.- From the setting of 3.7.1.4 the horizontal calibration knob shall be rotated from the median setting (coincidence position of left and right images) to plus 40 and minus 40. The range for each extreme shall be equivalent to the corresponding change in the number of UOE on the knob scale within 10 percent. In addition five UOE are allowed to cover the error in the positioning of the horizontal calibrating wedge relative to the ranging plane and its angular tolerance.

3.7.9.5 ICS calibration.- From the settings of 3.7.1.4 offset the ICS knob 20 UOE CCW, and re-establish reticle coincidence by rotation of the horizontal calibration knob. Then rotate the ICS knob 30 UOE CW and re-establish reticle coincidence by rotation of the horizontal calibration knob. The excursion of the horizontal calibration knob shall be 40 UOE plus or minus 10 UOE as measured on the knob scale.

3.7.10 Filters.- The introduction of filters into the optical system shall not cause more than 1 UOE change in horizontal calibration nor more than 0.1 mil in vertical calibration in coincidence.

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3.7.11 Compensator elevation error.- When the rangefinder is actuated to cover the range span from 500 to 4400 meters the elevation of the two target images shall not change with respect to each other by more than 0.1 mil.

3.7.12 Range accuracy.- With the rangefinder set as in 3.7.1.4, the indicating device for rotational position of the output shaft shall be set to the values computed from the formula: $P = 23.454 \text{ Log } 10^{R} - 58.577$ where "P" is the shaft position in turns and "R" is the target range in meters. At 1200 meters on the range scale, P= 13.64.

3.7.12.1 Output shaft excursion.- The range of rotation of the output shaft shall be from P= 3.76 plus or minus 0.05 to P= 27.25 plus or minus 0.20 turns, where the turn counter is adjusted in accordance with 3.7.12.

3.7.12.2 Output shaft tolerance.- When ranging on targets at the ranges specified in Table I the number of turns of the output shaft for each range shall not differ from the "P" value computed from the formula given in 3.7.12 in excess of the unit of error tolerances given in Table I. (see 6.8)

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TABLE I

Tolerance on Position of Rangefinder Output Shaft

Exact range in meters within spread	Range tolerance in UOE (Plus or Minus)
500 to 540	11
650 to 750	5
1100 to 1300	0
1600 to 2000	2
2200 to 2800	2
3800 to 4400	2

3.7.12.3 Range scale tolerance.- The readings of the range scale shall agree with the positions taken by the output shaft when checked to the requirements of Table II.

TABLE II

Tolerance on Reading of Range Scale

Position of output shaft	Reading of range scale	Tolerance on range scale reading (plus or minus)
520 meters	520 meters	5 meters
800 meters	800 meters	5 meters
1200 meters	1200 meters	0 meters
2000 meters	2000 meters	10 meters
4100 meters	4100 meters	20 meters

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3.7.13 Cross field errors.- From the median setting established in 3.7.1.4 and a 1200 (plus or minus 100 meters) target establish coincidence of the target with the horizontal calibration knob at a point 20 mils (plus or minus 2 mils) above below right and left of the center of the field (boresight cross). The median of 5 coincidence readings at each point shall not differ from the median UOE reading obtained at the center of the field by more than 6 UOE.

3.7.14 Base length.- The base length of the rangefinder measured optically, when identified production end housing assemblies are installed, shall be 79 inches plus or minus 0.15 inch.

3.7.15 Knob travel.- All knobs snall have sufficient movement to cover the entire range of their respective scales.

3.7.16 Boresight knobs.

3.7.16.1 Elevation.- Rotation of the elevation boresight knobs shall cause the line of sight to move at least 5 mils in depression and 2 mils in elevation from the nominal position established in 3.5.1(11) and(12).

3.7.16.2 Deflection.- Rotation of the deflection boresight knobs shall cause the line of sight to move at least 4.5 mils deflection each side of the nominal position established in 3.5.1(11) and (12).

3.7.16.3 Backlash.- Backlash in the boresight knobs shall not exceed 0.05 mil in the line of sight as read on the knob scales.

3.7.16.4 Locks.- Actuation of the locks of the deflection and elevation boresight knobs shall not cause more than 0.05 mil shift in the line of sight and not more than 0.05 mil rotation of the boresight knob, as measured on the knob scale.

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3.7.16.5 Lock retention.- When the locks of the boresight knobs are in the locked positions, the knob shall not move when a torque of 12 pound inches is applied to the knob.

3.7.17 Calibration knobs.

3.7.17.1 Backlash.- Backlash in the ICS knob shall not exceed 1/2 UOE and backlash in the horizontal calibration knob shall not exceed 1 UOE as read on the knob scales. Backlash in the halving knob as measured on the calibrating reticle shall not exceed 0.5 mils and backlash in the vertical calibration knob as read on the target reticle shall not exceed 0.25 mils.

3.7.17.2 Locks.- Locking of the horizontal and vertical calibration knobs by setting the knobs in the serration detent shall not introduce deflection error in excess of 1/4 UOE or elevation error in excess of the width of the index line, as read on the knob scales.

3.8 Operability.- When the rangefinder is exposed to the ambient temperatures indicated in Table III the maximum torque required for operation of the controls shall not exceed the values specified in the table.

3.9 Horizontal calibration.- Each rangefinder after vibration and after environmental requirements, prior to acceptance and shipment shall be checked for horizontal calibration readings on a target at 1200 meters plus or minus 100 meters and shall be made to meet the following mandatory requirements.

(a) Horizontal calibration of the target image shall be reset to be within "7" units of error (UOE) of the mid point of the total travel of the horizontal calibration knob.

(b) Using the ICS knob, a median of five ICS coincidence readings of the calibrating reticle shall read within 7 UOE of the (25) position of the ICS knob scale.

(c) The rangefinder shall meet the requirements of 3.7.2.1, 3.7.3, 3.10 and 3.10.1.

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TABLE III

Operation at Room and Environmental Temperatures

Control	Room +60° to 90°F (Standard Ambient Temperature)	Environmental -25° and 150°F	
Range Knob	70 oz. inches	16 lb. inches	70 oz. inches
Deflector (azimuth) and elevation boresight knobs	60 oz. inches	10 lb. inches	60 oz. inches
Horizontal and vertical calibration knobs (pull out force)	48 oz.	10 lb.	48 oz.
Horizontal and vertical calibration knob (rotation)	60 oz. inches	10 lb. inches	60 oz. inches
Halving knob	60 oz. inches	10 lb. inches	60 oz. inches
ICS knob	60 oz. inches	10 lb. inches	60 oz. inches
Diopter adjustment	30 oz. inches	10 lb. incnes	30 oz. inches
Rheostat knob	30 oz. inches	8 lb. inches	30 oz. inches
Light switches	*m.o.	m.o.	m.o.
Computer switch	m.o	m.o.	m.o.
Boresight and calibration locks	m.o.	m.o.	m.o.
Filter lever	m.o.	m.o.	m.o.
Occluder knob	m.o.	m.o.	m.o.

*m.o. - mechanically operable

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3.10 Sealing.- The rangefinder body and end housings shall show no evidence of leakage in excess of the following when subjected to an internal pressure of 6 inches of mercury at standard ambient temperature (plus 60 to 90 degrees Fahrenheit) for a period of twelve hours. Correction shall be made for any pressure variation resulting from temperature and barometric changes.

Rangefinder body	0.3 inch mercury
End housing	0.1 inch mercury

3.10.1 Flushing.- Upon satisfactory completion of the sealing requirements of 3.10, the rangefinder and the end housings shall be flushed with dry nitrogen. The flushing shall continue until the gaseous mixture bled from the rangefinder and the end housings have a dew-point of at least minus 25 degrees Fahrenheit. The instruments shall then be sealed with an internal pressure of dry nitrogen of between 0.2 and 0.5 inches of mercury.

3.11 Environmental requirements (see 3.9)

3.11.1 Storage temperatures.- The rangefinder with the identified end housings attached, shall be exposed to ambient temperatures of minus 80 and plus 160 degrees Fahrenheit until thermal equilibrium is reached. After thermal equilibrium has been re-established at standard ambient temperature, the rangefinder and end housings shall show no evidence of physical failure such as cement separation, bonding failure, glass breakage, casting failure, etc., and shall meet the requirements of 3.11. Readjustment of the rangefinder will be permitted following storage temperatures, but will require that the rangefinder be completely rechecked for conformance with all post vibration requirements of 3.7.1 to 3.7.17 inclusive.

3.11.2 Operational temperatures.- The rangefinder with the identified end housings attached shall be exposed to an ambient temperature of minus 40 degrees Fahrenheit until thermal equilibrium is reached, and inspected at that temperature to insure that all knobs, locks and operational adjustments are mechanically operable. The rangefinder with end housings attached shall then be exposed to an ambient temperature of minus 25 degrees Fahrenheit until thermal equilibrium is reached and then shall be inspected at this temperature for conformance with the

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requirements of 3.8 and Table III. The rangefinder with end housings attached shall be exposed to an ambient temperature of plus 150 degrees Fahrenheit until thermal equilibrium is reached and inspected at this temperature for conformance to the requirements of 3.8.

3.12 Post environmental check(See 3.9)

3.12.1 Optical check.- Any rangefinder which has been subjected to any part of the environmental requirements of 3.11 shall comply with the following requirements after thermal equilibrium has been re-established at standard ambient temperature.

(1) The line of sight of the left telescope through the boresight mark shall not have changed by more than 0.5 mil in deflection (azimuth) and 0.5 mil in elevation from the values in 3.7.1.1. The line of sight of the right telescope shall coincide with the aiming point of the left telescope within a maximum angular separation of 0.3 mil. Lack of horizontal coincidence of target image shall not exceed 40 UOE. Coincidence of the target images shall be re-established by using the horizontal calibration knob only. Set the horizontal calibration knob to the new median reading as obtained in preceding sentence; then, using the ICS knob, obtain a median from five ICS settings of the calibrating reticle. This median shall not differ by more than 5 UOE from median reading established in 3.7.1.4 prior to environmental exposure.

(2) The rangefinder shall meet the vertical calibration requirements of 3.7.1.2 and the requirements of 3.7.1.3

(3) The field error shall not have increased from the values established in 3.7.13 by more than 2 UOE.

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3.13 Sealing recheck.- Rangefinder and end housings which have met the environmental requirements of 3.11 shall be rechecked to the requirements of 3.10 for leakage and then be reflushed to meet the requirement of 3.10.1 failure of any rangefinder or end housing to meet the requirements of 3.10 after completion of the environmental requirements of 3.11 shall not be cause for rejection of the unit provided the rangefinder or end housing is resealed to meet the requirements of 3.10 and 3.10.1.

3.14 Electrical check.- Electrical checks may be conducted at any time prior to final acceptance of the rangefinder.

3.14.1 Insulation resistance.- Insulation resistance between ungrounded circuits and the body or frame and between isolated circuits shall not be less than 2 megohms with 500 volts dc applied.

3.14.2 Continuity.- The electrical circuits shall be checked for continuity and shall conform to the wiring diagram.

3.15 Identification marking.- Identification marking unless otherwise specified shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 General requirements.- Inspection shall be in accordance with General Specification MIL-I-45208.

4.2 Production model.- Unless otherwise specified in the contract, one rangefinder shall be selected by the Government inspector as soon as practicable after the start of production, and shall be inspected by the contractor under the surveillance of the Government. This inspection shall be conducted at the contractor's facility, or Government approved laboratory, using test equipment authorized for use on the contract. The model shall be inspected for all requirements of this specification by utilizing examination and testing methods prescribed herein.

4.2.1 Production model failure.- Should the production model fail to pass the prescribed tests, the contractor shall cease those operations which are causing the rejection and shall make the necessary corrections. When corrections in the process have been made, a new production model shall be examined and tested by the contractor under the surveillance of the Government. Acceptance of production lots will not commence until an acceptable production model is submitted and corrective actions have been taken to eliminate deficiencies.

4.3 Lot formation.- Inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.3.1 Defective units and lots shall be processed as specified in Standard MIL-STD-105.

4.4 Inspection of components. All positional and functional characteristics shown on drawings or contained in notes on the drawings or referenced specifications which indicate precision requirements and have a bearing on functional accuracy and performance of the rangefinder shall be inspected in accordance with the written inspection system required by Specification MIL-I-45208. These characteristics are defined as requirements affecting concentricity, squareness, parallelism, micro-finishes 46 RMS or better, threads, identification marking and performance.

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4.4.1 Sampling inspection.- Sampling inspection of components may be used in lieu of 100 percent inspection, and shall be in accordance with Table IV.

TABLE IV - SAMPLING PLAN

Lot Size Inclusive	Sample Size	Accept Lot	Reject Lot
0 - 25	Use 100% Inspection		
26 - 35	20	0	1
36 - 50	25	0	1
51 - 150	30	0	1
151 - 300	30	1	2
301 - over	60	2	3

4.4.2 Disposition of non-conforming product.- Acceptance and rejection numbers are expressed in terms of defective units. One or more defects on any unit shall constitute a defective unit. Disposition of non-conforming components shall be in accordance with Standard MIL-STD-105.

4.5 Material.- Control of material shall be in accordance with test data requirements of Specification MIL-I-45208.

4.6 Inspection equipment.- Supply, calibration, maintenance and use of inspection equipment shall be in accordance with Specification MIL-I-45208.

4.6.1 Accuracy requirements.- Inspection equipment which incorporates features for measuring values or sizes established by product requirements shall be capable of measuring those values within an accuracy of ten percent of the specified tolerance, except as otherwise specified. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

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4.6.2 Inspection equipment design requirements.

4.6.2.1 Device to test complete rangefinder (complete rangefinder is defined as depicted on Drawing F8205604).- This device shall provide for testing the following requirements: Orientation, calibration reticle setting accuracy, post vibration optical recheck, alinement of sighting reticles, image, coincidence reticle setting, parallax of sighting reticles, parallax of the calibrating reticles, parallax of the target images, accuracy of sighting reticles, plumb travel, parallelism of sighting reticle and image, parallelism of calibrating reticles, filters, compensator elevation error, range accuracy, field errors, base length, boresight knobs elevation, boresight elevation, boresight knobs deflection, boresight knobs backlash, boresight knobs locks, and post environmental optical check.

a. The device shall be equipped with suitable means to permit mounting the rangefinder, by means of the mounting bearings, in its normal operating position. The device shall be so constructed as to permit cross-leveling within 10 seconds and the horizontal mounting surface of the device shall be horizontal and parallel with respect to a datum line (theoretically) established by the mechanical centerline of the rangefinder within 25 seconds.

b. An appropriate target or targets, with acceptance and rejection limitation markings or pips marked thereon, utilized for measuring the reticle patterns shall be plumb and in line vertically and horizontally with the datum line within 25 seconds. The target or targets shall be so constructed as to be able to simulate target distance of from 500 meters and the left line of sight shall be 90 degrees with respect to the datum line within 30 seconds. The target lines of sight shall be parallel to each other in deflection within 1.5 seconds and parallel to each other in elevation within 10 seconds.

c. Provisions shall be made to permit rotation of the output shaft from 500 meters to 4400 meters as indicated on the turn counter, within an accuracy of plus or minus .05 turns.

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d. The device shall be equipped with a power supply capable of supplying the required power to the rangefinder during final acceptance testing.

e. An appropriate aperture plug, with an aperture diameter of .100 plus or minus .005, with a minimum movement of the aperture .030 to the left and .030 to the right, shall be provided. This aperture plug shall be utilized for testing parallax of the calibrating reticles and parallax of the target images.

4.6.2.2 Device to vibrate the rangefinder.- The device shall be a type capable of producing vertical motion at a constant frequency of thirty cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion). This device shall be equipped with an adapter which will permit the mounting of the rangefinder in its normal operating position. The base of the adapter shall be secured firmly to the device.

4.6.2.3 Device to test torque.- Torque wrench type testers with the following minimum ranges shall be utilized:

a. Zero to 16 pound inches, the accuracy of which shall be plus or minus 0.25 pound inches throughout the entire zone of excursion.

b. Zero to 70 ounce inches, the accuracy of which shall be plus or minus 0.25 ounce inches throughout the entire zone of excursion.

c. These devices shall be equipped with suitable adapters to accommodate testing range knob, deflection boresight knobs, elevation boresight knobs, halving knob, ICS knob, horizontal calibration knob, diopter adjustment and rheostat knob, and vertical calibration knob.

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4.7 Tests.

4.7.1 Classification of defects.- The classification of defects in Tables V and VI shall constitute the minimum inspection requirements to be performed prior to acceptance or rejection by lots. The samples for test listed in Table VI shall be selected from the samples that have successfully passed the test listed in Table V. The Government reserves the right to inspect for any or all applicable requirements and to reject individual non-conforming items.

4.7.2 Disposition of rejected lots.- Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lots shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

TABLE V - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 0.65%	Req't Par.	QA. Par.	Test Method
101. Horizontal calibration knob travel, pre-vibration	3.5.2.1	4.8.1.1.1	DST-Visual
102. Vertical calibration knob travel, pre vibration	3.5.2.2	4.8.1.1.2	DST-Visual
103. Azimuth and elevation boresight knobs, pre-vibration	3.5.2.3	4.8.1.1.3	DST-Visual
104. Calibration reticle setting accuracy, pre-vibration	3.5.3	4.8.1.2	DST-Visual
105. Physical failure, post-vibration	3.7.1	4.8.2.1	Visual-Tactile

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TABLE V - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 0.65%	Req't Par.	QA. Par.	Test Method
106. Cleanliness post-vibration	3.7.2	4.8.2.2	Visual
107. Alinement of sighting reticles	3.7.1.1	4.8.3.1	DST-Visual
108. Coincidence of target images	3.7.1.2	4.8.3.2	DST-Visual
109. Calibration reticle setting	3.7.1.3	4.8.3.3	DST-Visual
110. Horizontal calibration	3.7.1.4	4.8.3.4	DST-Visual
111. Bright spots	3.7.2.2	4.8.3.5	Visual
112. Parallax of sighting reticles	3.7.3.1	4.8.3.6	DST-Visual
113. Parallax of calibration reticles	3.7.3.2	4.8.3.7	DST-Visual
114. Parallax of target images	3.7.3.3	4.8.3.8	DST-Visual
115. Exit pupil	3.7.4	4.8.3.9	Standard Measuring Equipment (SME)
116. Eyepiece focus	3.7.5	4.8.3.10	SME-Visual
117. Definition	3.7.6	4.8.3.11	SME-Visual
118. Light transmission	3.7.7	4.8.3.12	SME-Visual
119. Accuracy of sighting reticles	3.7.8.1	4.8.3.13.1	DST-Visual
120. Plumb travel	3.7.8.2	4.8.3.13.2	DST-Visual
121. Parallelism of sighting reticle and image	3.7.8.3	4.8.3.13.3	DST-Visual
122. Tilt variation between reticles	3.7.8.4	4.8.3.13.4	DST-Visual
123. Parallelism of calibrating reticle	3.7.9.1	4.8.3.14.1	DST-Visual
124. Reversing reticles	3.7.9.2	4.8.3.14.2	DST-Visual
125. Vertical movement accuracy	3.7.9.3	4.8.3.14.3	DST-Visual
126. Horizontal calibration	3.7.9.4	4.8.3.14.4	DST-Visual
127. ICS calibration	3.7.9.5	4.8.3.14.5	DST-Visual
128. Filter error	3.7.10	4.8.3.15	DST-Visual
129. Compensator elevation error	3.7.11	4.8.3.16	DST-Visual
130. Range accuracy	3.7.12	4.8.3.17	DST-Visual

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TABLE V - CLASSIFICATION OF DEFECTS (Cont'd)

MAJOR: AQL 0.65%	Req't Par.	QA. Par.	Test Method
131. Cross field error	3.7.13	4.8.3.18	DST-Visual
132. Base length	3.7.14	4.8.3.19	DST-Visual
133. Knob travel	3.7.15	4.8.3.20	Visual-Tactile
134. Elevation boresight knob movement	3.7.16.1	4.8.3.21.1	DST-Visual
135. Deflection (azimuth) boresight knob movement	3.7.16.2	4.8.3.21.2	DST-Visual
136. Backlash, boresight knobs	3.7.16.3	4.8.3.21.3	DST-Visual
137. Boresight lock error	3.7.16.4	4.8.3.21.4	DST-Visual
138. Boresight knob lock retention	3.7.16.5	4.8.3.21.5	DST-Visual
139. Backlash, ICS and horizontal calibration knob	3.7.17.1	4.8.3.22.1	DST-Visual
140. Backlash, halving and vertical calibration knob	3.7.17.1	4.8.3.22.2	DST-Visual
141. Calibration lock error	3.7.17.2	4.8.3.22.3	DST-Visual
142. Operability, controls	3.8	4.8.4	DST
143. Sealing, main housing and end housing	3.10	4.8.5	DST
144. Insulation resistance	3.14.1	4.8.6	SME
145. Continuity	3.14.2	4.8.7	SME
146. Identification marking	3.15	4.8.13	Visual

MINOR: NONE DEFINED

NOTE: The above tests shall be conducted at standard ambient temperature (plus 60 to plus 90 degrees Fahrenheit) in accordance with the quality assurance paragraph (QA. Par.) listed.

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TABLE VI - CLASSIFICATION OF DEFECTS

Use Inspection Level L7 and Sampling Plan Table IV-A of Standard MIL-STD-105

CRITICAL: NONE DEFINED

MAJOR: AQL 6.5%	Req't Par.	QA. Par.	Test Method
101. Controls, operability (minus 40°F)	3.11.2	4.8.8.1	Visual-Tactile
102. Controls, torque (minus 25°F, plus 150°F)	3.8	4.8.8.2	Device, Special Testing (DST)
103. Physical failure, rangefinder	3.10.1	4.8.8.3	Visual-Tactile
104. Alinement of sighting reticles	3.12.1(1)	4.8.9.1	DST-Visual
105. Image coincidence	3.12.1(1)	4.8.9.2	DST-Visual
106. Horizontal calibration	3.12.1(1)	4.8.9.3	DST-Visual
107. Vertical calibration of images	3.12.1(2)	4.8.9.4	DST-Visual
108. Calibration reticles setting	3.12.1(2)	4.8.9.5	DST-Visual
109. Field error	3.12.1(3)	4.8.9.6	DST-Visual
110. Sealing, main housing and end housing	3.13	4.8.10	DST-Visual

MINOR: NONE DEFINED

NOTE: The tests for characteristics 101 and 102 shall be conducted at the temperatures indicated. The tests for characteristics 103 thru 110 inclusive shall be conducted at standard ambient temperature (plus 60 to plus 90 degrees Fahrenheit) subsequent to the temperature cycling test. All tests shall be conducted in accordance with the quality assurance paragraph (QA. Par.) listed.

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4.8 Test methods and procedures.

4.8.1 Orientation of rangefinder for tests, previbration.- The requirements for orientation shall be met with the rangefinder mounted on special test equipment with a line through the centers of the 2 mounting bearings horizontal. The geometric centers of the 2 mounting bearings and the bore on Adapter Assembly 8201530 shall be in a plane perpendicular to the horizontal line within the tolerance specified on Drawing F8205604. A line of sight through the rangefinder shall be directed to a calibrated target, positioned at the specified distance, appropriate for measuring the accuracy requirements of the rangefinder. The horizontal target line shall be coincident to the line through the centers of the 2 mounting bearings. The vertical target line shall be perpendicular to that line through the centers of the mounting bearings.

4.8.1.1 Knob travel-previbration.

4.8.1.1.1 Horizontal calibration knob.- When the horizontal calibration knob has been set to the median coincidence reading (coincidence of left and right target image) established in 3.5.1(9), the knob shall be at the mid-point of rotation within the tolerance specified. Rotation in UOE displaced from initial established mid-point shall be read from knob scale. Slip scale to aline zero index to fixed index upon successful completion of test.

4.8.1.1.2 Vertical calibration knob.- When vertical coincidence of the left and right target image has been established as specified in 3.5.1(8), the vertical calibration knob shall be at the mid-point of rotation within the tolerance specified. Rotation from initial established mid-point shall be the measured distance between the fixed and movable index lines. Slip scale to aline index lines upon successful completion of test. Mark position of knob for subsequent test.

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4.8.1.1.3 Deflection and elevation boresight knobs.- The deflection and elevation boresight knobs shall be set to their mid-point of travel as specified in 3.5.1(10), and the scales slipped to aline the deflection three (3) and elevation two (2) graduation line to the fixed index. When the line of sight through the boresight mark (center of boresight cross) of the left and right sighting reticle is aligned to the positions specified in 3.5.1(11) and 3.5.1(12), the deflection and elevation boresight knobs shall be at their mid-point of rotation within the tolerance specified, as measured on the boresight scales. Slip scale to aline the deflection three (3) and elevation two (2) graduation line to the fixed indices upon successful completion of test.

4.8.1.2 Calibration reticle setting accuracy, previbration.- With the rangefinder oriented as in 3.5.1 and the calibration knobs set to mid-point within the tolerance specified, the lower calibration reticle shall meet the requirements specified as measured on a target appropriate for measuring displacement of the calibration reticle with respect to the sighting reticle.

4.8.1.2.1 The upper calibration reticle shall be adjusted in elevation by means of the halving knob to meet the requirements specified. Mark position of halving knob for subsequent test.

4.8.1.2.2 The ICS knob shall move the upper calibration reticle in azimuth to vertically aline the calibration reticles. A group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob. The median of this group of settings shall not exceed the tolerance specified as read on the ICS knob scale.

4.8.2 Vibration.- The rangefinder shall be mounted in its normal operating position upon special test equipment and vibrated in accordance with the requirements specified.

4.8.2.1 Physical failure, post vibration.- As a result of vibration, the rangefinder shall show no signs of physical failure of mechanical or optical parts.

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4.8.2.2 Cleanliness, post-vibration.- The rangefinder shall be inspected for cleanliness in accordance with Specification MIL-O-13830, and shall meet the requirements specified.

4.8.3 Rangefinder, post-vibration.- Subsequent to vibration, the rangefinder, shall be set up as specified in 4.8.1. Certain tests shall require auxiliary testing equipment to be used in conjunction with the special test equipment specified in 4.8.1 for determination of compliance with the requirements.

4.8.3.1 Alinement of sighting reticles.- The line of sight through the boresight mark of the left and right sighting reticles shall again be alined to the positions specified in 3.5.1(11) and 3.5.1(12). Any change in the alinement of the sighting reticles from the values established in 4.8.1.1.3 shall not exceed the tolerance specified, as read on the knob scale.

4.8.3.2 Coincidence of target image.- When the vertical calibration knob is rotated to the position established in 4.8.1.1.2 vertical coincidence of the target images shall not have changed by more than the tolerance specified, as read on a target appropriate for measuring displacement of target images. The median of five readings to bring the target images into horizontal coincidence, by means of the horizontal calibration knob, shall not differ from the zero setting established in 4.8.1.1.1 by more than the tolerance specified. Any change in UOE shall be read on the horizontal calibration knob scale. Slip scale to the new zero upon successful completion of test.

4.8.3.3 Calibration reticle setting.- Subsequent to vibration, the space between the two calibration reticles shall meet the requirements specified when the vertical calibration knob is rotated to the position established in 4.8.1.1.2 and the halving knob is rotated to the position established in 4.8.1.2.1.

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4.8.3.4 Horizontal calibration.- The horizontal calibration knob shall be set to the median reading obtained in 4.8.3.2. A group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob. The difference between the median of this group and the value established in 4.8.1.2.2 shall not exceed the tolerance specified.

4.8.3.5 Bright spots.- The reticles, illuminated by the lamps in the rangefinder, shall be visually inspected for conformance with the requirements specified.

4.8.3.6 Parallax of sighting reticles.- The test shall be conducted with each sighting reticle projected into the optical system, in turn. Parallax at the reticle boresight mark for each optical system shall not exceed the tolerance specified. When the image of the left and right sighting reticles are superimposed, the parallax between the reticles shall not exceed the tolerance specified. Parallax is any apparent movement of the reticles in relation to each other or to the target in the field of view caused by movement of the observer's head.

4.8.3.7 Parallax of the calibration reticles.- The test shall be conducted with the aid of an auxiliary testing device containing a central aperture that will permit controlled scanning of the exit pupil. This device shall be positioned over the rangefinder eyepiece. A group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob when the eye is positioned the specified distance to the left of exit pupil center. A group of five readings shall be taken with the eye positioned to the right of exit pupil center. The difference between the median of the left group and the median of the right group shall not exceed the tolerance specified, as read on the ICS knob scale.

4.8.3.8 Parallax of the target images.- The auxiliary testing device described in the test for parallax of calibration reticles shall be used for this test. The images of a target at 1200 meters plus or minus 100 meters shall be brought into coincidence by means of the range knob. Coincidence shall then be disturbed by rotation of the horizontal calibration knob. A group of five horizontal coincidence settings of the target image shall be accomplished by means of the horizontal calibration knob when the eye is positioned the specified distance to the left of exit pupil center. A group of five readings shall be taken with the eye positioned to the right of exit pupil center. The difference between the median of the left group and the median of the right group shall not exceed the tolerance specified, as read on the horizontal calibration knob scale.

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4.8.3.9 Exit pupil.- This test shall be conducted in accordance with Specification MIL-O-13830 and shall meet the requirements specified.

4.8.3.10 Eyepiece focus.- The diopter movement of the eyepiece shall be tested manually and visually for the requirements specified. Focus of the rangefinder eyepiece shall be tested with the aid of a calibrated dioptometer, having a magnification of at least three-power (3X), adjusted for infinity focus for the observer's eye, and the diopter scale set to zero. The zero setting of the rangefinder eyepiece diopter scale shall be within the tolerance specified when sharp focus of the left target image is obtained.

4.8.3.11 Resolution.- This test shall be conducted in accordance with Specification MIL-O-13830. The resolving power of each system shall meet the requirements specified. The auxiliary telescope used in this test shall be at least a three-power (3X) instrument.

4.8.3.12 Light transmission.- Filters shall be transferred out of the optical systems for this test. Light rays from a concentrated source entering the rangefinder, through each end housing window, shall be measured at the eyepiece by means of a photometer, consisting of a light source, photocell and micrometer. The difference in light transmission between the two optical systems shall meet the requirements specified.

4.8.3.13 Sighting reticles.

4.8.3.13.1 Accuracy of sighting reticles.- This test shall be conducted in accordance with Specification MIL-O-13830. The angle subtended by the total horizontal lines and spaces of the left and right sighting reticles shall meet the requirements specified.

4.8.3.13.2 Plumb travel.- A line of sight through the boresight mark of the left sighting reticle coincident to a plumb line shall not deviate from that plumb line by more than the tolerance specified when the rangefinder is manually elevated from 8 degrees depression to 19 degrees elevation.

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4.8.3.13.2.1 Rangefinder movement.- Rangefinder shall be elevated and depressed to meet the requirements specified.

4.8.3.13.3 Parallelism of sighting reticle and image.- Parallelism of the longest horizontal reticle line to the image of a horizontal target line shall be measured on a target appropriate for measuring any cant of the reticle line. Parallelism of the left and right sighting reticle shall meet the requirements specified.

4.8.3.13.4 Tilt variation between reticles.- When the right sighting reticle is superimposed on the left sighting reticle, the tilt variation between reticles, as measured at one extremity of the 40 mil reticle line, shall not exceed the tolerance specified when the other extremity of both 40 mil reticle lines are coincident. Tilt between reticles shall be measured on a target appropriate for measuring tilt variation between reticles within the tolerance specified.

4.8.3.14 Coincident reticles.

4.8.3.14.1 Parallelism of Calibrating reticles.- Each calibrating reticle, in turn, shall be tested for parallelism to the image of a plumb line and shall be parallel within the tolerance specified. When one extremity of a vertical side of the calibration reticle is alined coincident to the image of a plumb line, the other extremity shall be parallel to that plumb line within the width of the calibration reticle at that extremity.

4.8.3.14.2 Reversing reticles. With the calibration reticles in reversed position (upper bar immediately under the lower bar), a group of five horizontal coincidence settings of the calibration reticles shall be accomplished by means of the ICS knob. The difference between the median of this group and the value established in 4.8.3.4 shall not exceed the tolerance specified as read on the ICS knob scale. Upon successful completion of the vertical movement accuracy test, restore the calibration reticles to their proper position.

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4.8.3.14.3 Vertical movement accuracy.- Set the horizontal calibration knob and the ICS knob to the median values obtained in 4.8.3.4. The vertical calibration knob shall be rotated to the maximum CCW position and the upper calibration reticle, controlled by the halving knob, shall be positioned immediately above the lower reticle (separated by a minimum distinguishable spacing). A group of five horizontal coincidence settings of the calibration reticle shall be accomplished by means of the ICS knob. The vertical calibration knob shall then be rotated to the maximum CW position and the upper calibration reticle again positioned as above. A group of five horizontal coincidence settings of these reticles shall be accomplished by means of the ICS knob. The difference between the median of the first group and the median of the second group shall not exceed the tolerance specified as read on the ICS knob scale. Upon successful completion of this test, return calibration reticles to normal position.

4.8.3.14.4 Horizontal calibration.- The line of sight of the rangefinder shall be directed to a target of known distance within 1100 and 1300 meters and range scale set to agree with the range chosen. The target shall be in coincidence and the horizontal calibration and ICS knobs set to established median positions. The horizontal calibration knob shall be rotated plus 40 UOE, then coincidence of the target images re-established by means of the ranging knob. The corresponding change in range, as read on the range scale, shall be equivalent to the change in the number of UOE within the tolerance specified. This test shall be repeated with the horizontal calibration knob rotated to minus 40 UOE.

For computing the change in range that corresponds to the change in UOE use the following approximation:

$$\text{The change in range for 1 UOE} = 2.90 \left(\frac{R}{1000} \right)^2$$

Where R = Range scale reading

This approximation is good for a maximum of 10 UOE.

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4.8.3.14.5 ICS calibration.- The horizontal calibration knob and the ICS knob shall be set to the values established in 4.8.3.4. The ICS and horizontal calibration knobs shall be rotated to the positions specified, as measured on the knob scales. The position of the horizontal calibration knob, as a result of this excursion, shall not exceed the tolerance specified, as read on the horizontal calibration knob scale.

4.8.3.15 Filters.- Introduction of the filters shall not cause the target images to shift horizontally with respect to each other by more than the tolerance specified.

4.8.3.16 Compensator elevation error.- The rangefinder range scale shall be set to 500 meters with the target images in vertical coincidence. Any vertical shift of the target images shall not exceed the tolerance specified when the range scale is actuated from 500 meters to 4400 meters. Image shift shall be measured on a target appropriate for measuring image shift within the specified tolerance.

4.8.3.17 Range accuracy.- The horizontal calibration knob shall be set to the value established in 4.8.3.4. An auxiliary testing device that shall be capable of accurately indicating rotational positions of the rangefinder output shaft within the tolerances specified in Table I, when the range scale is set to any range between 500 meters and 4400 meters shall be attached to the rangefinder output shaft. This testing device shall indicate an output shaft rotational value (P) of 13.64 when the range scale indicates 1200 meters of range.

4.8.3.17.1 The range of rotation of the output shaft shall meet the requirements specified as measured on the testing device.

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4.8.3.17.2 Output shaft tolerance.- The rangefinder line of sight shall be directed to a target or targets positioned at known distances between 500 - 540 meters, 650 - 750 meters, 1100 - 1300 meters, 2200 - 2800 meters and 3800 - 4400 meters. The "P" value for the known range shall be computed from the formula specified and shall be computed for each target. The median of each group of five range readings (rotational position of output shaft) shall meet the requirements within the UOE tolerance specified in Table I. Unit of error tolerances shall be converted to output shaft rotations using the formula 1 UOE equals 0.000029R and then compared with the difference between the indicated output shaft value of the median range reading and the computed "P" value for a specific range.

4.8.3.17.3 Range scale tolerance.- When the rangefinder output shaft is rotated to the nominal positions (P-value) specified for the ranges in column I, Table II, the range scale shall indicate these ranges respectively within the tolerance specified. The nominal positions of the output shaft shall be computed from the formula $P = 23.454 \text{ Log } 10_R - 58.577$.

4.8.3.18 Cross field error.- This test shall require five targets positioned at 1200 meters plus or minus 100 meters. One target shall be positioned at the center of the field (a line of sight through the reticle boresight mark) with the remaining four targets positioned at the specified number of mils from the center of the field. Set the rangefinder range scale to the range chosen and sight on target in the center of the field. The target at the center of the field shall be in coincidence when the horizontal calibration and ICS knobs are set to their established median positions. A group of five horizontal coincidence settings shall be taken on each target, in turn, by means of the horizontal calibration knob. The difference between the median of the center target and the medians of each group of five settings shall not exceed the tolerance specified as measured on the horizontal calibration knob scale.

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4.8.3.19 Base length.- All control knobs of the rangefinder shall be set to their individual established median positions and the range scale set to 4400 meters of range. A plumb line shall be positioned approximately 3 to 4 inches in front of each end housing window. The measured distance between plumb lines shall be 79.000 inches plus or minus 0.15 inches (2.0059 meters plus or minus 3.81 millimeters). The test for base length shall be performed with the aid of an auxiliary testing device, positioned at the eyepiece of the rangefinder. This device shall be capable of focusing on the etched dots on Objectives 8200012, 8200005, and the plumb lines. The scale or reticle of the device shall be capable of measuring the actual separation between dot images and plumb line images formed at the rangefinder eyepiece. Focus the device on the etched dots, then turn occluder of rangefinder so that left dot image is observed. Record scale reading of position of left dot. Turn occluder so that right dot image is observed. Record scale reading of position of right dot. The difference between the two recorded readings shall be used as the "A" factor in the formula for computing base length error. Repeat the operations listed above to measure the difference between plumb lines. The difference between the two recorded readings shall be used as the "B" factor in the following formula for computing base length error where 10 equals the power of the rangefinder.

$$10\ (A-B) = \text{Base length error}$$

The base length error shall not exceed the tolerance specified.

4.8.3.20 Knob travel.- All knobs shall be visually and manually inspected for conformance to the requirement specified.

4.8.3.21 Boresight knobs.

4.8.3.21.1 Elevation.- The lines of sight, through the boresight mark of the left and right sighting reticle, shall be oriented to the position established in 4.8.1.1.3. Each mil of rotation of the elevation boresight knobs, clockwise and counterclockwise from the established nominal position, shall cause a comparable change in the line of sight and shall meet the excursion requirements specified as measured on a target appropriate for measuring elevation movement of the reticle.

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4.8.3.21.2 Deflection.- The line of sight, through the boresight mark of the left and right sighting reticle, shall be oriented to the position established in 4.8.1.1.3. Each mil of rotation of the deflection boresight knobs, clockwise and counterclockwise from the established nominal position, shall cause a comparable change in the line of sight and shall meet the excursion requirements specified as measured on a target appropriate for measuring deflection movement of the reticle.

4.8.3.21.3 Backlash boresight knobs.- Rotate the deflection boresight knob in a continuous clockwise direction to move the vertical reticle line of the boresight reticle into coincidence with a vertical target line. Record the setting of the knob scale. Rotate the knob through one full resolution clockwise to move the vertical reticle line past the vertical target line, then by a continuous counterclockwise rotation of the azimuth knob bring the vertical reticle line into coincidence with the vertical target line and read the knob scale. The difference between the recorded reading and the second reading is the amount of backlash. Repeat this operation for the elevation boresight knob using the horizontal reticle line and a horizontal target line. Backlash for each elevation and each deflection knob shall not exceed the limit specified. Rotation of the knobs must be uniform in one direction. The reticle line shall not override the target line during these operations. If override occurs, this test is nullified and the operations specified will again be performed.

3.8.3.21.4 Boresight locks.- Actuation of the deflection and elevation boresight knob locks shall not cause the line of sight to shift nor the boresight knobs to rotate more than the tolerance specified.

4.8.3.21.5 Boresight knob lock retention.- This test shall be performed with special testing device affixed to the boresight knobs that shall be capable of measuring applied torques. The boresight knob lock retention shall meet the requirements specified when the indicated torque is applied to the knobs.

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4.8.3.22 Calibration knobs.

4.8.3.22.1 Backlash in ICS knob and horizontal calibration knob.- These tests shall be conducted similar to 4.8.3.21.3. The upper calibration reticle shall be alined to vertical target line by means of the ICS knob. The clockwise rotation of the knob shall be to the clockwise stop before beginning the counterclockwise rotation. The lower calibration reticle shall be alined to the vertical target line by means of the horizontal calibration knob. The results of this test shall meet the requirements specified.

4.8.3.22.2 Backlash in halving knob and vertical calibration knob.- These tests shall be conducted similar to 4.8.3.21.3. The upper calibration reticle shall be alined to a horizontal target line by means of the halving knob. The lower calibration reticle shall be alined to the horizontal target line by means of the vertical calibration knob. The results of this test shall meet the requirements specified.

4.8.3.22.3 Calibration locks.- This test may be conducted concurrent with the backlash test for calibration knobs. Any error caused by the engagement of the horizontal and vertical calibration knobs with the serration detents shall not exceed the tolerance specified.

4.8.4 Operability.- The maximum torques required to rotate the controls of the rangefinder shall not exceed those values specified in column I, Table III, while at an ambient temperature of plus 60 to 90 degrees Fahrenheit. These tests shall be conducted with special testing devices capable of measuring the applied torques.

4.8.5 Sealing and flushing of rangefinder main body and end housing assembly.

4.8.5.1 The rangefinder body and end housing assembly shall be flushed with dry nitrogen. The rangefinder body and end housing assembly shall then be sealed with an internal pressure of 6 inches of mercury. After a period of 12 hours, the body and end housing shall show no evidence of leakage of dry nitrogen in excess of the tolerance specified. The sealing test shall be conducted with special test equipment that shall be capable of measuring any drop in internal pressure of rangefinder body and end housing assembly within the tolerance specified.

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4.8.5.2 Upon satisfactory completion of the sealing test, the rangefinder body and end housing assemblies shall be bled to the internal pressure specified. At this pressure, the dewpoint temperature shall meet the requirements specified as measured on a dewpoint tester.

4.8.6 Insulation resistance.- This test shall be conducted in accordance with Specification MIL-F-13926 by use of standard measuring equipment and shall meet the requirements of 3.14.1. Remove all lamp bulbs and remove ground connection, identified as wire number 17 on Drawing C8620852 before conducting this test.

4.8.7 Continuity.- The electrical circuits shall be checked for continuity by use of standard measuring equipment and shall conform to the wiring diagram.

4.8.8 Environmental test of rangefinder.- Environmental test shall be performed using the contractor's Government approved refrigeration and heating equipment. The operability and torque test on the rangefinder may be conducted concurrent with the storage test. Caution must be exercised so as not to expose the rangefinder to thermal shock during environmental test. Temperature changes shall be gradual.

4.8.8.1 Controls, operability.- When thermal equilibrium is reached at minus 40 degrees Fahrenheit, the controls listed in Table III shall be mechanically operable.

4.8.8.2 Controls, torque.- When thermal equilibrium is reached at minus 25 and plus 150 degrees Fahrenheit, all controls shall meet the applicable torque requirements.

4.8.8.3 Physical failures, post-environmental.- Subsequent to the exposure to the storage and operational temperatures, the rangefinder shall be returned to room temperature and examined for evidence of physical failure of optical and mechanical parts. Failure of rangefinder to meet this test shall be cause for rejection of the rangefinder.

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4.8.9 Post-environmental test of rangefinder.- Subsequent to the test specified in 4.8.8.3 the rangefinder shall be set up as specified in 4.8.1. The boresight knobs and calibration knobs shall be set to their established values.

4.8.9.1 Alinement of sighting reticles.- The line of sight through the boresight mark of the left sighting reticle shall meet the requirements specified. The line of sight through the boresight mark of right sighting reticle shall coincide with the aiming point of the left telescope within the tolerance specified.

4.8.9.2 Image coincidence.- Obtain a new median of five coincidence settings of the target images by means of the horizontal calibration knob. The difference between the new median reading and the median obtained in 4.8.3.2 shall not exceed the UOE tolerance specified.

4.8.9.3 Horizontal calibration.- Set horizontal calibration known to median setting obtained in 4.8.9.2. Obtain a new median of five ICS settings of the calibrating reticle using the ICS knob. The difference between the new median setting and the median obtained in 4.8.3.4 shall not exceed the UOE tolerance specified.

4.8.9.4 Vertical calibration of images.- This test shall be conducted as indicated in 4.8.3.2. Vertical calibration of the images shall meet the requirements specified.

4.8.9.5 Calibration reticles setting.- This test shall be conducted as indicated in 4.8.3.3. The calibrating reticles setting shall meet the requirements specified.

4.8.9.6 Field error.- This test shall be conducted as indicated in 4.8.3.18. Field error shall meet the requirements specified.

4.8.10 Post-environmental, sealing recheck of the rangefinder and end housing assemblies.- Subsequent to environmental test, the rangefinder and end housing assemblies subjected to that test shall again meet the requirements specified. This test shall be as specified in 4.8.5. Failure of any rangefinder or end housing to meet the requirements shall not be cause for rejection provided the rangefinder is resealed and flushed to meet the requirements specified. Disassembly of the rangefinder to reseal shall not be permitted.

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4.8.11 Horizontal calibration, post-vibration, post-environmental.- Each rangefinder that has been adjusted to meet the requirements of 3.9 shall be subjected to the following test prior to final acceptance and shipment:

(a) The rangefinder shall be set-up as specified in 4.8.1 and tested in accordance with 4.8.1.1.1. Horizontal calibration of the target images shall meet the requirements specified.

(b) A group of five coincidence settings of the calibration reticles shall be made by means of the ICS knob. The median of the five settings shall not exceed the tolerance specified as read on the ICS scale.

(c) The rangefinder shall be inspected in accordance with 4.8.2.2 and shall meet the requirements specified.

(d) The rangefinder shall be tested in accordance with 4.8.3.6, 4.8.3.7 and 4.8.3.8 and shall meet the requirements specified.

(e) The rangefinder shall be tested in accordance with 4.8.5 and shall meet the requirements specified.

4.8.12 Final acceptance.- Final acceptance of the lot of rangefinders (with identified end housings) shall be withheld by the Government inspector until the sample lots of rangefinders (with identified end housings), representative of that production lot has satisfactorily passed the applicable requirements of this detailed specification. The rangefinder with identified end housing shall be shipped as a complete unit.

4.8.13 Identification markings.- Marking shall be visually inspected for conformance to Standard MIL-STD-130.

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4.8.14 Packaging, packing and marking.- Quality Assurance Provisions for packaging, packing and marking shall be in accordance with Specification MIL-P-14232.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- The rangefinder shall be packaged, packed and marked in accordance with Specification MIL-F-14619/P5800850. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The Rangefinder: XM20 is a full field superimposed coincidence type instrument, intended for installation in the 90 MM Gun, Tank: M48A3. This rangefinder is commander operated. It employs a 10 power optical system, with a 4 degree field of view and has a base length, including end housings of 79 inches and furnishes range data to a trajectory computing system.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Selection of an applicable level of preservation, packaging and packing.
(c) Applicable stock number
(d) Provisions for production model including date of delivery (see 3.1).

6.3 Design information.- Information regarding specific design not covered or referenced by this specification may be obtained from the procuring activity upon specific request of the contractor.

6.4 Position of adapter link assembly.- No rangefinder shall be accepted for testing which has not been previously inspected, prior to assembly of the main hanger bearings, for precise positioning of the Adapter Link Assembly 8201530 as required by Drawing F8620777-4. Two production end housings shall be identified as components of each rangefinder by stenciling the rangefinder serial number, and end housing position (left and right) on the exterior to each end housing. Under no circumstances will temporary substitution of other end housings be permitted for completion of requirements herein or for shipment of the rangefinder.

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6.5 Post vibration.- Should the rangefinder fail to meet all requirements of 3.7.1, the contractor may request that the identified end housing assemblies be removed and replaced, using only the same identified end housings, in order to insure that the installation of the end housings was not at fault. Should the results of the second installation be at variance with the first, one or more additional installations may be made until two consecutive installations repeat within required limits, after which the instrument shall be considered acceptable to this stage (3.7.1).

6.6 Definitions.

6.6.1 Median.- The term "median" as used herein, is defined as that range or setting which divides the series into an upper and lower half. Example - in a set of five range readings 2000, 2050, 2080, 2060, 2025 the median would be 2050. Two ranges are higher and two are lower.

6.6.2 Standard measuring equipment (SME).- Standard measuring equipment includes the common hand type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement.

6.6.3 Special testing device(DST).- Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

6.7 Environmental exposure.- Pre-Ordnance environmental exposure by the contractor for the purpose of stabilizing a complete instrument or any part thereof, prior to Ordnance Inspection, will be permitted and encouraged provided such treatment is part of the established process for all instruments. This process shall not be limited to those instruments selected for environmental testing.

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6.8 Output shaft.- For the computation of UOE with respect to the amount of turns of the output shaft use the relationship that 0.000029R (R is range in meters) corresponds to 1 UOE change in ranging. (This formula should not be used for application for values in excess of 5 UOE).

NOTICE: When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

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